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Technical Specification

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5G Security Assurance Specification (SCAS);
User Plane Function (UPF)
(Release 19)

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Foreword

This Technical Specification has been produced by the 3rd Generation Partnership Project (3GPP).

The contents of the present document are subject to continuing work within the TSG and may change following formal TSG approval. Should the TSG modify the contents of the present document, it will be re-released by the TSG with an identifying change of release date and an increase in version number as follows:

Version x.y.z

where:

x the first digit:

1 presented to TSG for information;

2 presented to TSG for approval;

3 or greater indicates TSG approved document under change control.

Y the second digit is incremented for all changes of substance, i.e. technical enhancements, corrections, updates, etc.

z the third digit is incremented when editorial only changes have been incorporated in the document.

In the present document, modal verbs have the following meanings:

shall indicates a mandatory requirement to do something

shall not indicates an interdiction (prohibition) to do something

The constructions "shall" and "shall not" are confined to the context of normative provisions, and do not appear in Technical Reports.

The constructions "must" and "must not" are not used as substitutes for "shall" and "shall not". Their use is avoided insofar as possible, and they are not used in a normative context except in a direct citation from an external, referenced, non-3GPP document, or so as to maintain continuity of style when extending or modifying the provisions of such a referenced document.

should indicates a recommendation to do something

should not indicates a recommendation not to do something

may indicates permission to do something

need not indicates permission not to do something

The construction "may not" is ambiguous and is not used in normative elements. The unambiguous constructions "might not" or "shall not" are used instead, depending upon the meaning intended.

can indicates that something is possible

cannot indicates that something is impossible

The constructions "can" and "cannot" are not substitutes for "may" and "need not".

will indicates that something is certain or expected to happen as a result of action taken by an agency the behaviour of which is outside the scope of the present document

will not indicates that something is certain or expected not to happen as a result of action taken by an agency the behaviour of which is outside the scope of the present document

might indicates a likelihood that something will happen as a result of action taken by some agency the behaviour of which is outside the scope of the present document

might not indicates a likelihood that something will not happen as a result of action taken by some agency the behaviour of which is outside the scope of the present document

In addition:

is (or any other verb in the indicative mood) indicates a statement of fact

is not (or any other negative verb in the indicative mood) indicates a statement of fact

The constructions "is" and "is not" do not indicate requirements.

1 Scope

The present document contains requirements and test cases that are specific to the UPF network product class. It refers to the Catalogue of General Security Assurance Requirements and formulates specific adaptions of the requirements and test cases. It also specifies the requirements and test cases unique to the UPF network product class.

2 References

The following documents contain provisions which, through reference in this text, constitute provisions of the present document.

- References are either specific (identified by date of publication, edition number, version number, etc.) or non-specific.
- For a specific reference, subsequent revisions do not apply.
- For a non-specific reference, the latest version applies. In the case of a reference to a 3GPP document (including a GSM document), a non-specific reference implicitly refers to the latest version of that document *in the same Release as the present document*.

[1] 3GPP TR 21.905: "Vocabulary for 3GPP Specifications".

[2] 3GPP TS 33.501: "Security architecture and procedures for 5G system".

[3] 3GPP TS 33.117: "Catalogue of general security assurance requirements".

[4] 3GPP TS 23.501: "System Architecture for 5G system".

[5] 3GPP TS 29.281: "General Packet Radio System (GPRS) Tunnelling Protocol User Plane (GTPv1-U) ".

[6] 3GPP TS 23.060: "General Packet Radio Service (GPRS); Service description; Stage 2".

[7] 3GPP TR 33.926: "Security Assurance Specification (SCAS) threats and critical assets in 3GPP network product classes".

[8] Void

3 Definitions of terms, symbols and abbreviations

3.1 Terms

For the purposes of the present document, the terms and definitions given in 3GPP TR 21.905 [1] and the following apply. A term defined in the present document takes precedence over the definition of the same term, if any, in 3GPP TR 21.905 [1].

3.2 Symbols

Void.

3.3 Abbreviations

For the purposes of the present document, the abbreviations given in 3GPP TR 21.905 [1] and the following apply. An abbreviation defined in the present document takes precedence over the definition of the same abbreviation, if any, in 3GPP TR 21.905 [1].

Void.

4 UPF-specific security requirements and related test cases

4.1 Introduction

The present document describes the following security requirements and the related test cases for UPF:

- Security functional requirements and the related test cases (clause 4.2),
- Adaptations of hardening requirements and the related test cases (clause 4.3), and
- Adaptations of basic vulnerability testing requirements and the related test cases (clause 4.4).

The above categories are aligned with those specified in TS 33.117 [3]. The text on pre-requisites for testing in clause 4.1.2 of TS 33.117 [3] applies also to the present document.

4.2 UPF-specific security functional requirements and related test cases

4.2.1 Introduction

The security functional requirements and the related test cases specific for UPF are described in the clause.

4.2.2 Security functional requirements on the UPF deriving from 3GPP specifications and related test cases

4.2.2.0 General

The general approach in TS 33.117 [3] clause 4.2.2.1 apply to the UPF network product class. The requirements and test cases in TS 33.117 [3] clause 4.2.2.2 related to SBA/SBI aspect are not applicable.

4.2.2.1 Confidentiality protection of user data transported over N3 interface

Requirement Name: Confidentiality protection of user data transported over N3 interface.

Requirement Reference: TS 33.501 [2], Clause 9.3

Requirement Description: The transported user data between gNB and UPF is confidentiality protected as specified in TS 33.501 [2], clause 9.3.

Threat Reference: TR 33.926 [7], Clause L.2.2, "No protection or weak protection for user plane data ".

TEST CASE:

NOTE 1: This test case is only applicable to UPF supporting IPSec in N3 interface without the use of a SEG.

Test Name: TC_UP_DATA_CONF_UPF

Purpose:

Verify that the transported user data between gNB and UPF are confidentiality protected over N3 interface.

Procedure and execution steps:

Pre-Condition:

- UPF network product is connected in simulated/real network environment.
- The tunnel mode IPsec ESP and IKE certificate authentication is implemented.
- Tester shall have knowledge of the security parameters of tunnel for decrypting the ESP packets.

- Tester shall have access to the N3 interface between gNB and UPF.
- Tester shall have knowledge of the confidentiality algorithm and confidentiality protection keys used for encrypting the encapsulated payload.

Execution Steps:

The requirement mentioned in this clause is tested in accordance with the procedure mentioned in clause 4.2.3.2.4 of TS 33.117 [3].

Expected Results:

The user data transported between gNB and UPF is confidentiality protected.

Expected format of evidence:

Evidence suitable for the interface, e.g., evidence can be presented in the form of screenshot/screen-capture.

4.2.2.2 Integrity protection of user data transported over N3 interface

Requirement Name: Integrity protection of user data transported over N3 interface.

Requirement Reference: TS 33.501 [2], Clause 9.3

Requirement Description: The transported user data between gNB and UPF is integrity protected as specified in TS 33.501 [2], clause 9.3.

Threat Reference: TR 33.926 [7], Clause L.2.2, "No protection or weak protection for user plane data"

TEST CASE:

NOTE 1: This test case is only applicable to UPF supporting IPSec in N3 interface without the use of a SEG

Test Name: TC_UP_DATA_INT_UPF

Purpose:

Verify that the transported user data between gNB and UPF are integrity protected over N3 interface.

Procedure and execution steps:

Pre-Condition:

- UPF network product is connected in simulated/real network environment.
- The tunnel mode IPsec ESP and IKE certificate authentication is implemented.
- Tester shall have knowledge of the security parameters of tunnel for decrypting the Encapsulated Security Payload (ESP) packets.
- Tester shall have knowledge of the authentication algorithm (Hash Message Authentication Code) and the protection keys.

Execution Steps:

The requirement mentioned in this clause is tested in accordance to the procedure mentioned in clause 4.2.3.2.4 of TS 33.117 [3].

Expected Results:

The user data transported between gNB and UPF is integrity protected.

Expected format of evidence:

Evidence suitable for the interface, e.g., evidence can be presented in the form of screenshot/screen-capture.

4.2.2.3 Replay protection of user data transported over N3 interface

Requirement Name: Replay protection of user data transported over N3 interface

Requirement Reference: TS 33.501 [2], Clause 9.3

Requirement Description: The transported user data between gNB and UPF is replay protected as specified in TS 33.501, clause 9.3.

Threat Reference: TR 33.926 [7], Clause L.2.2, "No protection or weak protection for user plane data"

TEST CASE:

NOTE 1: This test case is only applicable to UPF supporting IPSec in N3 interface without the use of a SEG.

Test Name: TC_UP_DATA_REPLAY_UPF

Purpose:

Verify that the transported user data between gNB and UPF are replay protected.

Procedure and execution steps:

The following procedure is executed if UPF supports IPsec.

Pre-Condition:

- UPF network product is connected in simulated/real network environment.
- The tunnel mode IPsec ESP and IKE certificate authentication is implemented.
- Tester shall have knowledge of the security parameters of tunnel for decrypting the ESP packets.
- Tester shall have access to the original user data transported via N3 reference point between gNB and UPF.

Execution Steps:

The requirement mentioned in this clause is tested in accordance with the procedure mentioned in clause 4.2.3.2.4 of TS 33.117 [3].

Expected Results:

The user data transported between UE and UPF is replay protected.

Expected format of evidence:

Evidence suitable for the interface, e.g., evidence can be presented in the form of screenshot/screen-capture.

4.2.2.4 Protection of user data transported over N9 interface Within a PLMN

Requirement Name: Protection of user data transported over N9 within a PLMN.

Requirement Reference: TS 33.501 [2], Clause 9.9

Requirement Description: As specified in clause 9.9 in TS 33.501 [2], interfaces internal to the 5G Core can be used to transport signalling data as well as privacy sensitive material, such as user and subscription data, or other parameters, such as security keys. Therefore, confidentiality and integrity protection is required.

For the protection of the non-SBA internal interfaces, such as N4 and N9, NDS/IP is used as specified in TS 33.501 [2], clause 9.9 .

Threat Reference: TR 33.926 [7], Clause L.2.2, "No protection or weak protection for user plane data "

TEST CASE:

NOTE 1: This test case is only applicable to UPF supporting IPSec in N3 interface without the use of a SEG.

Test Name: TC_UP_DATA_CONF_UPF_N9

Purpose:

Verify that the protection mechanism implemented for user data transport over N9 interface in a PLMN conforms to the selected security profile.

Procedure and execution steps:

Pre-Condition:

- UPF network products are connected in simulated/real network environment.
- The tunnel mode IPsec ESP and IKE certificate authentication is implemented.
- Tester shall have knowledge of the security parameters of tunnel for decrypting the ESP packets.
- Tester shall have access to the N9 interface between two UPFs within a PLMN.
- Tester shall have knowledge of the confidentiality algorithm and confidentiality protection keys used for encrypting the encapsulated payload.

Execution Steps:

The requirement mentioned in this clause is tested in accordance with the procedure mentioned in clause 4.2.3.2.4 of TS 33.117 [3].

Expected Results:

The user data transported on N9 within a PLMN is protected.

Expected format of evidence:

Evidence suitable for the interface, e.g., evidence can be presented in the form of screenshot/screen-capture.

4.2.2.5 Signalling Data Protection

Requirement Name: Protection of signalling data transported over N4 interface.

Requirement Reference: TS 33.501 [2], Clause 9.9

Requirement Description: As specified in clause 9.9 in TS 33.501 [2], interfaces internal to the 5G Core can be used to transport signalling data as well as privacy sensitive material, such as user and subscription data, or other parameters, such as security keys. Therefore, confidentiality and integrity protection is required.

For the protection of the non-SBA internal interfaces, such as N4 and N9, NDS/IP is used as specified in TS 33.501 [2], clause 9.9.

Threat Reference: TR 33.926 [7], Clause L.2.3, "No protection or weak protection for signalling data over N4 interface"

TEST CASE:

NOTE 1: This test case is only applicable to UPF supporting IPSec in N3 interface without the use of a SEG.

Test Name: TC_CP_DATA_CONF_UPF_N4

Purpose:

Verify that the protection mechanism implemented for signalling data transmitted over N4 conforms to selected security profile.

Procedure and execution steps:

Pre-Condition:

- UPF and SMF network products are connected in simulated/real network environment.
- The tunnel mode IPsec ESP and IKE certificate authentication is implemented.
- Tester shall have knowledge of the security parameters of tunnel for decrypting the ESP packets.

- Tester shall have access to the N4 interface between SMF and UPF.
- Tester shall have knowledge of the confidentiality algorithm and confidentiality protection keys used for encrypting the encapsulated payload.

Execution Steps:

The requirement mentioned in this clause is tested in accordance with the procedure mentioned in clause 4.2.3.2.4 of TS 33.117 [3].

Expected Results:

The signalling data transported over N4 interface is protected.

Expected format of evidence:

Evidence suitable for the interface, e.g., evidence can be presented in the form of screenshot/screen-capture.

4.2.2.6 TEID uniqueness

Requirement Name: TEID uniqueness.

Requirement Reference:

TS 23.501 [4], Clause 5.8.2.3.1; TS 29.281 [5], Clause 5.1; TS 23.060 [6], Clause 14.6

Requirement Description:

Allocation and release of CN Tunnel Info is performed when a new PDU Session is established or released. This functionality is supported either by SMF or UPF, based on operator's configuration on the SMF as specified in TS 23.501[4], clause 5.8.2.3.1.

Tunnel Endpoint Identifier (TEID): This field unambiguously identifies a tunnel endpoint in the receiving GTP U protocol entity. The receiving end side of a GTP tunnel locally assigns the TEID value the transmitting side has to use as specified in TS 29.281[5], clause 5.1.

The TEID is a unique identifier within one IP address of a logical node as specified in TS 23.060 [6], clause 14.6.

Threat Reference: TR 33.926 [7], Clause L.2.4, "Failure to assign unique TEID for a session"

TEST CASE:

Test Name: TC_TEID_ID_UNIQUENESS_UPF

Purpose:

Verify that the TEID generated by UPF under test for each new GTP tunnel is unique.

Pre-Conditions:

Test environment is set up with SMF, which may be real or simulated, and UPF under test. The tester is able to trace traffic between the UPF under test and the SMF (real or simulated). SMF configures UPF under test to generate the TEIDs.

Execution Steps:

1) The tester intercepts the traffic between the UPF under test and the SMF.
2) The tester triggers the maximum number of concurrent N4 session establishment requests.
3) The tester captures the N4 session establishment responses sent from UPF to SMF and verifies that the F-TEID created for each generated response is unique.

Expected Results:

The F-TEID set in each different N4 session establishment response is unique.

Expected format of evidence:

Files containing the triggered PFCP messages (e.g. pcap trace).

4.2.2.7 IPUPS

Requirement Name: IPUPS packeting handling

Requirement Reference: TS 33.501[2], clause 5.9.3.4

Requirement Description:

The IPUPS only forwards GTP-U packets that contain an F-TEID that belongs to an active PDU session and discard all others as specified in TS 33.501 [2], clause 5.9.3.4.

Threat Reference: TR 33.926 [7], Clause L.2.5, "invalid user plane data forwarding"

TEST CASE:

NOTE 1: This test case is only applicable to UPF supporting IPUPS.

Test Name: TC_IPUPS_PACKET_HANDLING

Purpose:

Verify that the packets not belonging to an active PDU session is discarded.

Pre-Conditions:

Test environment is set up with a V-SMF, an H-SMF, an H-UPF and a gNB which may be simulated.

Execution Steps:

1) The V-SMF requests the UPF with IPUPS functionality under test to establish an N4 session for a PDU session in home-routing roaming. The UPF with IPUPS functionality under test responds to the SMF with the F-TEID for the N9 tunnel towards the H-UPF, and the F-TEID for the N3 tunnel towards the gNB.

2) The V-SMF requests the H-SMF to establish a PDU session providing the received F-TEID for the N9 tunnel.

3) The H-SMF requests the H-UPF to establish an N4 session providing the received F-TEID for the N9 tunnel. H-UPF in the response provides its F-TEID for the N9 tunnel. The H-SMF provides the received F-TEID from the H-UPF to the V-SMF.

4) The V-SMF requests the gNB to allocate resource for the PDU session providing the F-TEID for the N3 tunnel received at step 1. The gNB replies with its F-TEID for the N3 tunnel to the V-SMF.

5) The V-SMF provides the UPF with IPUPS functionality under test with the received F-TEID assigned by the gNB for the N3 tunnel and the received F-TEID assigned by the H-UPF for the N9 tunnel.

6) The H-UPF is triggered to send GTP-U packets using the F-TEID assigned by the V-UPF for the N9 tunnel.

7) The H-UPF is triggered to send GTP-U packets using an F-TEID different than the one assigned by V-UPF for N9 tunnel.

Expected Results:

When the H-UPF is triggered to send GTP-U packets using the F-TEID assigned by the V-UPF for the N9 tunnel (step 6 in the execution steps), GTP-U packets are witnessed over the N3 tunnel.

When the H-UPF is triggered to send GTP-U packets using an F-TEID different than the one assigned by the V-UPF (step 7 in the execution steps), no GTP-U packets are witnessed over the N3 tunnel.

Expected format of evidence:

Files recording the GTP packets captured (e.g. pcap trace).

4.2.2.8 Protection against malformed GTP-U messages

Requirement Name: Protection against malformed GTP-U messages

Requirement Reference: TS 33.501[2], clause 5.9.3.4

Requirement Description: The IPUPS discards malformed GTP-U messages as specified in TS 33.501[2], clause 5.9.3.4.

Threat Reference: TR 33.926 [7], Clause L.2.6, "Threats of malformed GTP-U messages"

TEST CASE:

NOTE 1: This test case is only applicable to UPF supporting IPUPS.

Test Name: TC_IPUPS_MALFORED_MESSAGES

Purpose:

Verify that malformed messages are discarded by UPF.

Pre-Conditions:

The pre-conditions in clause 4.4.4 of TS 33.117 [3] apply, except that fuzzing tools supporting GTP-U protocol is available.

Execution Steps:

The execution steps follow those in clause 4.4.4 of TS 33.117 [3], except that the protocol the fuzzing tool is executed against is GTP-U and the interface is N9.

Expected Results:

The expected results in clause 4.4.4 of TS 33.117 [3] apply except that the protocol and the interface contained in the testing documentation are GTP-U and N9 respectively.

Expected format of evidence:

The expected format of evidence in clause 4.4.4 of TS 33.117 [3] apply.

4.2.3 Technical baseline

4.2.3.1 Introduction

The present clause provides baseline technical requirements.

4.2.3.2 Protecting data and information

4.2.3.2.1 Protecting data and information – general

There are no UPF-specific additions to clause 4.2.3.2.1 of TS 33.117 [3].

4.2.3.2.2 Protecting data and information – unauthorized viewing

There are no UPF-specific additions to clause 4.2.3.2.2 of TS 33.117 [3].

4.2.3.2.3 Protecting data and information in storage

There are no UPF-specific additions to clause 4.2.3.2.3 of TS 33.117 [3].

4.2.3.2.4 Protecting data and information in transfer

There are no UPF-specific additions to clause 4.2.3.2.4 of TS 33.117 [3].

4.2.3.2.5 Logging access to personal data

There are no UPF-specific additions to clause 4.2.3.2.5 of TS 33.117 [3].

4.2.3.3 Protecting availability and integrity

There are no UPF-specific additions to clause 4.2.3.3 of TS 33.117 [3].

4.2.3.4 Authentication and authorization

There are no UPF-specific additions to clause 4.2.3.4 of TS 33.117 [3].

4.2.3.5 Protecting sessions

There are no UPF-specific additions to clause 4.2.3.5 of TS 33.117 [3].

4.2.3.6 Logging

There are no UPF-specific additions to clause 4.2.3.6 of TS 33.117 [3].

4.2.4 Operating systems

There are no UPF-specific additions to clause 4.2.4 of TS 33.117 [3].

4.2.5 Web Servers

There are no UPF-specific additions to clause 4.2.5 of TS 33.117 [3].

4.2.6 Network Devices

There are no UPF-specific additions to clause 4.2.6 in TS 33.117 [3].

4.3 UPF-specific adaptations of hardening requirements and related test cases

4.3.1 Introduction

This clause specifies the UPF-specific adaptations of hardening requirements and related test cases.

4.3.2 Technical baseline

There are no UPF-specific additions to clause 4.3.2 in TS 33.117 [3].

4.3.3 Operating systems

There are no UPF-specific additions to clause 4.3.3 in TS 33.117 [3].

4.3.4 Web servers

There are no UPF-specific additions to clause 4.3.4 in TS 33.117 [3].

4.3.5 Network devices

There are no UPF-specific additions to clause 4.3.5 in TS 33.117 [3].

4.3.6 Network functions in service-based architecture

There are no UPF-specific additions to clause 4.3.6 in TS 33.117 [3].

4.4 UPF-specific adaptations of basic vulnerability testing requirements and related test cases

4.4.1 Introduction

There are no UPF specific additions to clause 4.4.1 of TS 33.117 [3].

4.4.2 Port Scanning

There are no UPF specific additions to clause 4.4.2 of TS 33.117 [3].

4.4.3 Vulnerability scanning

There are no UPF specific additions to clause 4.4.3 of TS 33.117 [3].

4.4.4 Robustness and fuzz testing

The test cases under clause 4.4.4 of TS 33.117 [3] are applicable to UPF.

The interfaces defined for the UPF are in clause 4.2.3 of TS 23.501 [4].

According to clause 4.4.4 of TS 33.117 [3], the transport protocols available on the interfaces providing IP-based protocols need to be robustness tested. Following TCP/IP layer model and considering all the protocols over transport layer, for UPF, the following interfaces and protocols are in the scope of the testing:

- For N3: the UDP and GTP-U protocols.
- For N4: the UDP and PFCP protocols.
- For N9: the UDP and GTP-U protocols.
- The N6 is the connection with the Data Network (DN). The protocols used in this interface are not defined by the 3GPP and are not under the scope of the present document.

NOTE: There could be other interfaces and/or protocols requiring testing under clause 4.4.4 of TS 33.117 [3].

Annex A (informative): Change history

Change history							
Date	**Meeting**	**Tdoc**	**CR**	**Rev**	**Cat**	**Subject/Comment**	**New version**
2019-10						EditHelp review, editorial changes	16.0.1
2019-12	SA#86	SP-191138	0002	1	F	Corrections for clean-up and alignment	16.1.0
2020-12	SA#90e	SP-201004	0003	-	F	Reference of general SBA/SBI aspect in 33.513	16.2.0
2021-09	SA#93e	SP-210848	0005	-	B	New test cases of IPUPS to TS 33.513	17.0.0
2022-12	SA#98e	SP-221157	0009	-	F	Correction of requirement references in UPF test case	17.1.0
2023-06	SA#100	SP-230615	0012	1	A	Correction of SBA test for UPF	17.2.0
2023-06	SA#100	SP-230604	0010	1	F	SCAS release reference corrections	18.0.0
2023-06	SA#100	SP-230604	0013	-	B	Changes for SCAS UPF for Rel18	18.0.0
2023-12	SA#102	SP-231339	0014	1	F	Correction of protocol in Expected format of evidence	18.1.0
2025-07	SA#108	SP-250657	0015	-	F	Correction of test names and clean up of 33.513	19.0.0